

Lab 6

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Lab 6

Lab 6: Creating an Android app to implement Retrofit with JSON

1. Add the following dependencies to the build.gradle.kts file.

```
implementation("androidx.compose.runtime:runtime-livedata:1.5.0-alpha01")
implementation("androidx.lifecycle:lifecycle-viewmodel-compose:2.8.0")
implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.6.1")
implementation("androidx.compose.material3:material3:1.2.0")
implementation("androidx.compose.foundation:foundation:1.6.0")
// Retrofit & Gson
implementation("com.squareup.retrofit2:retrofit:2.9.0")
implementation("com.squareup.retrofit2:converter-gson:2.9.0")
```

2. Create a data class named "Post".

```
data class Post(
    val userId: Int,
    val id: Int,
    val title: String,
    val body: String
)
```

3. Define an interface named "ApiService".

```
import retrofit2.Retrofit
import retrofit2.converter.gson.GsonConverterFactory
import retrofit2.http.GET

interface ApiService {
    @GET("posts")
    suspend fun getPosts(): List<Post>
}

object RetrofitClient {
    val api: ApiService by lazy {
        Retrofit.Builder()
            .baseUrl("https://jsonplaceholder.typicode.com/")
            .addConverterFactory(GsonConverterFactory.create())
            .build()
            .create(ApiService::class.java)
    }
}
```

4. Define a view model class named "PostViewModel".

```
package com.vinay.retrofitjsonapp
import androidx.lifecycle.ViewModel
import androidx.lifecycle.viewModelScope
import kotlinx.coroutines.flow.MutableStateFlow
import kotlinx.coroutines.flow.StateFlow
import kotlinx.coroutines.launch

class PostViewModel : ViewModel() {
    private val _posts = MutableStateFlow<List<Post>>(emptyList())
```

```

import androidx.lifecycle.ViewModel
import androidx.lifecycle.viewModelScope
import kotlinx.coroutines.flow.MutableStateFlow
import kotlinx.coroutines.flow.StateFlow
import kotlinx.coroutines.launch

class PostViewModel : ViewModel() {
    private val _posts = MutableStateFlow<List<Post>>(emptyList())
    val posts: StateFlow<List<Post>> = _posts

    init {
        viewModelScope.launch {
            try {
                _posts.value = RetrofitClient.api.getPosts()
            } catch (e: Exception) {
                // Handle error
            }
        }
    }
}

```

5. Add the following code to the MainActivity.kt file.

```

package com.vinay.retrofitjsonapp

import ads_mobile_sdk.h6
import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.Column
import androidx.compose.foundation.layout.Spacer
import androidx.compose.foundation.layout.fillMaxWidth
import androidx.compose.foundation.layout.height
import androidx.compose.foundation.layout.padding
import androidx.compose.foundation.lazy.LazyColumn
import androidx.compose.foundation.lazy.items
import androidx.compose.material3.Card
import androidx.compose.material3.ExperimentalMaterial3Api
import androidx.compose.material3.MaterialTheme
import androidx.compose.material3.Scaffold
import androidx.compose.material3.Text
import androidx.compose.material3.TopAppBar
import androidx.compose.runtime.Composable
import androidx.compose.runtime.collectAsState
import androidx.compose.runtime.getValue
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp

import androidx.lifecycle.viewmodel.compose.viewModel

import com.vinay.retrofitjsonapp.ui.theme.RetrofitJsonAppTheme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            RetrofitJsonAppTheme {
                PostScreen()
            }
        }
    }
}

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun PostScreen(postViewModel: PostViewModel = viewModel()) {
    val posts by postViewModel.posts.collectAsState()

    Scaffold(
        topBar = {
            TopAppBar(
                title = { Text("Posts") }
            )
        }
    )
}

```

```

        RetrofitJsonAppTheme {
            PostScreen()
        }
    }
}

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun PostScreen(postViewModel: PostViewModel = viewModel()) {
    val posts by postViewModel.posts.collectAsState()

    Scaffold(
        topBar = {
            TopAppBar(
                title = { Text("Posts") }
            )
        }
    ) { paddingValues ->
        LazyColumn(modifier = Modifier
            .padding(paddingValues)
            .padding(16.dp)
        ) {
            items(posts) { post ->
                PostItem(post)
            }
        }
    }
}

@Composable
fun PostItem(post: Post) {
    Card(
        modifier = Modifier
            .fillMaxWidth()
            .padding(vertical = 8.dp)
    ) {
        Column(modifier = Modifier.padding(16.dp)) {
            Text(
                text = post.title,
                style = MaterialTheme.typography.titleMedium
            )
            Spacer(modifier = Modifier.height(4.dp))
            Text(
                text = post.body,
                style = MaterialTheme.typography.bodySmall
            )
        }
    }
}

```

6. Check if everything seems good and run the app.